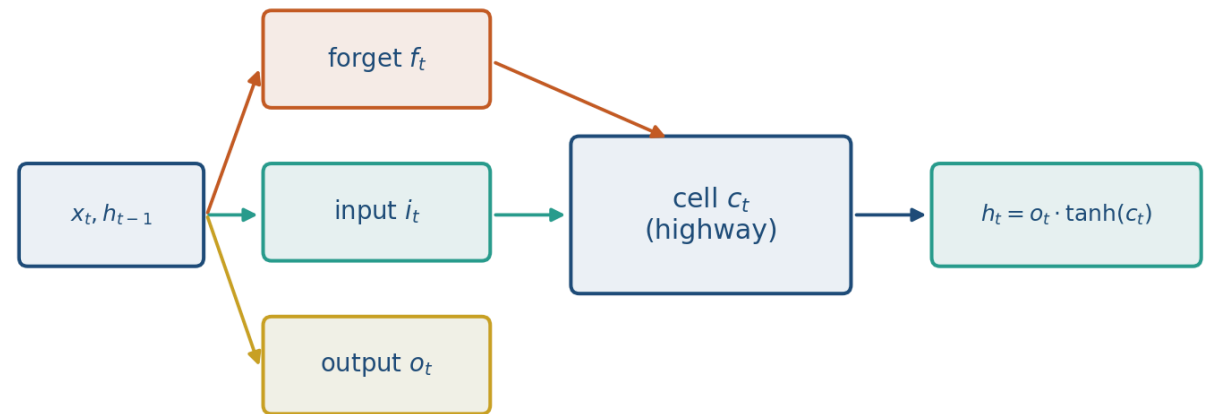


# LSTM: a highway plus three gates

- Forget, input, output. The cell can copy.
- If  $f_t \approx 1$  and  $i_t \approx 0$ , memory persists.

LSTM: a cell highway plus learned gates



# GRU: two gates

- Reset and update. The hidden state is the memory.
- Still a loop: better than vanilla, worse than attention at long range.

GRU: two gates, no separate cell state

